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A model for gear life with surface and subsurface survival: Tribological effects

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ABSTRACT

The paper describes a model for gear fatigue life based on the well-established methods used in dynamic load capacity of machine components. The model applies the Weibull's weakest link of material strength and the Lundberg-Palmgren theory for bearing dynamic load ratings. Significant, of the present model, is the introduction of a well-defined separation between the risk of surface initiated failures and the traditional subsurface fatigue of the gear contact. This opens new possibilities for the use of specialized endurance models to predict the surface life of the gear contact. A further advantage of the proposed model is the use of a statistical definition of the fatigue life of gears. Current design practices for gears are condition based rules. Essentially it is required to satisfy a fatigue limit stress which is further reduced using service penalty factors. Contrary to this approach, the proposed model makes use of a L10 life associated to a 90% reliability for the gear population. Comparison between the experimentally obtained gear endurances and L10 predicted life, using the present theory, indicates the ability of the model to account for the gear endurance and the contribution of the surface to the performance. This new approach is felt to be of advantage for future progress in gear design.

1. Introduction

Selection and design methods for gears are in general based on the Lewis equation [1] for the bending of beams. AGMA standard [2] introduces additional safety factors into the Lewis equation to account for stress concentration in the tooth root, overload and load distribution on the tooth. In addition to the fatigue resistance of the tooth root, AGMA also includes the verification of the maximum stress condition developed at the very surface of the tooth contact.

AGMA introduces an equation for the surface durability based on “dry” contact stresses, so that when the contact stress exceeds a certain critical value, it is considered that the gear teeth will pit. These methods have been successfully used to design and size gears since several decades back. However, with the increasing industrial demand to downsize machine components, reduce energy consumption and manufacturing costs, engineers are constantly looking for more up-to-date ways to optimise the dimensions of mechanical components.

The latest tribology knowledge in automatic lubrication, advanced surface finish and cleanliness control can clearly contribute to the gear durability. This can offer new opportunities to reduce over-design and increase mechanical efficiency. Today, within gear standard organisations, considerable efforts are made to assess gear micropitting risk [3], this indicates the importance of this topic for the durability of gears. However, up to now, micropitting risk can only indirectly be considered in the gear design process and, contrary to the approach proposed in this paper, no specific guidelines are provided with regard to the life expectancy of the gear.

On the contrary, rolling bearings have taken advantage, through the years, of unique and specialized modelling features for the prediction of the expected fatigue life of the rolling contact in relation to the operating conditions. The most significant are:

- the application of Weibull's weakest link theory of material strength, [4], which takes into consideration the material stress volume at risk.
- the use of Lundberg-Palmgren dynamic load rating theory, [5,6].
- statistical approach in the definition of the fatigue endurance, i.e. 90% reliability for the L10 life.
- The application of Ioannides-Harris [7] fatigue stress criterion allowing the inclusion of the effect of the lubrication quality and cleanliness in the rolling contact fatigue life.

Moreover, very recently a new general approach for rolling contact fatigue life was introduced. In this new method the surface-originated

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Nomenclature

A	Damage risk area m^2
A_1	Proportionality constant in life-stress equation [-]
$\overline{A}$	Proportionality constant in subsurface life-stress equation [-]
$\widehat{A}$	Constant in Eq. (9) [-]
a_u	Fatigue limit load-life modifying factor [-]
$\overline{B}$	Proportionality constant in surface life-stress equation [-]
b	Hertzian semi-width along meshing direction m
c	Exponent in the gear life equation [-]
e	Exponent in the gear life equation (standardized Weibull slope) [-]
E'	Combined elasticity modulus of the contact (EHL) Pa
f_e	Effective flack width m
$f_1 ... f_n$	Constants in Eq. (6) [-]
h	Exponent in the gear life equation [-]
I_s	Surface damage integral [-]
$\widetilde{I}_s$	Normalized surface damage integral, Eq. (6) [-]
k, k'	Constants, Eq. (A7) and Eq. (C2) [-]
k_0, k'_0	Material constants, Eq. (A9) and Eq. (C2) [-]
k_1, k'_1	Material constants, Eq. (A9) and Eq. (C2) [-]
L	Rating life M revs
L_{10}, L_{50}	Rating life with 90%, and respectively 50% reliability M revs
l_e	Effective involute length in the loaded zone m
N	Life in number of load cycles [-]
N_1	Number of teeth in pinion [-]
N_2	Number of teeth in gear [-]
p	Exponent in the gear life equation, Eq. (A10) [-]
p_o	Maximum Hertzian contact pressure Pa
Q	Normal contact load on the gear tooth N
Q_u	Normal fatigue limit load of the tooth N
Q_c	Dynamic load capacity for the tooth N
Q_m	Dynamic load capacity for meshing pair (normal) N
R	Equivalent contact radius of the bodies m
R_a	Arithmetical mean deviation of the surface m
R_q	Root mean squared of the surface, $R_q \approx 1.25 R_a m$
S	Reliability [-]
S	Sliding to rolling ratio, $S = \frac{2(u_2 - u_1)}{u_1 + u_2}$ [-]
S_R	Surface fatigue indicator [-]
T	Proportionality constant linking p_o and τ_o [-]
u	Conversion factor from gear revolutions to load cycles in a tooth [-]
u_1	Surface 1 speed m/s
u_2	Surface 2 speed m/s
V	Stressed volume m^3
W	Tangential load on the gear tooth N
W_u	Tangential component of the fatigue limit load of the tooth N
W_m	Dynamic load capacity for meshing pair (tangential) N
w	Exponent in the gear life equation, Eq. (A15) [-]
z	Depth from the surface for the maximum stress amplitude in the stress-based model m
z_o	Depth from the surface for the maximum orthogonal shear stress m

Greek Symbols

β	Shape parameter, i.e. slope, of the Weibull distribution [-]
η	Environmental factor, $\eta = \eta_b \eta_c$ [-]
η_b	Lubrication factor [-]
η_c	Contamination factor [-]
κ	Lubrication quality in rolling bearings (see ISO 281) [-]
σ	Stress amplitude in the stress-based model Pa
σ_u	Fatigue limit in the stress-based model Pa
φ	Pressure angle Deg
ψ	Helical angle Deg
Ψ_{gear}	Constant in Eq. (9) [-]
τ_{xz}	Orthogonal shear stress Pa
τ_o	Maximum orthogonal shear stress amplitude Pa
τ_u	Fatigue limit (orthogonal shear stress amplitude) Pa

Subscripts

1,2	Refers to pinion and gear, respectively [-]
	Refers to surface and subsurface, respectively [-]
v, s	Refers to the stressed volume and stressed area [-]

damage is explicitly formulated into the basic fatigue equations of the rolling contact. Considering the survival probability of the surface as a failure risk that is separated from the Hertzian stresses of the subsurface, [8,9]. This opens new possibilities for the use of specialized tribology models to describe surface related failures of the rolling contact.

Gears have not yet taken advantage of these modelling concepts. Neither the usage of L10 or of dynamic load rating has really taken off in gear design. This despite the significant efforts during the seventies, due to the pioneering work of Coy et al. [10–13].

However, there are few examples of cross-fertilisation between bearing and gears modelling. Recently one of the authors of this paper, applied a surface distress (micropitting) model, originally developed for rolling bearings, to study spur gears micropitting [14]. The surface distress model that was used, included the computation of the concurrent effects of fatigue and mild wear. The same model was used in [8] to derive a surface distress function part of a generalized rolling bearing life model.

The objective of the current paper is to present an adaptation of the model [8], to study its potential application to gears. See Fig. 1 the principal concept, about the present model illustrated. The stress areas of the surface and subsurface of the gear contact are analysed separately regarding their contribution to the survival risk of the gear mesh.

This might help to clarify the advantages of this life rating method in comparison to the current ones, and will present the possibilities offered by the use of a more integrated approach for the life of the contacting surfaces of gears. For the time being, the current methodology will not include the tooth root stresses, but in the future an adaptation could be developed. Thus, the Lewis equation will remain valid for that aspect of the gear design. The model is presented as a concept, since calibration with actual gear test was not attempted and remains very limited at this stage.

2. The dynamic load capacity of gears

Coy et al. [10–13] have developed the equations corresponding to the dynamic capacity of spur and helical gears and will not be repeated here. Instead a quick review is carried out in a more general way to arrive to the same equations in Appendix A.

Therefore by considering Eqs. (A14) and (A15) from Appendix A, the dynamic load capacity for any pair of meshing gears has the generic equation:

$$W_m = \left[N_1\left(\frac{1}{Q_{c1}}\right)^w + N_2\left(\frac{1}{Q_{c2}}\right)^w\right]^{-1/w} \cos(\varphi) \tag{1}$$

with Q_{c1} and Q_{c2} being the individual tooth dynamic load capacities for the pinion and the gear and they are given by Eq. (A9) with $u = 1$.

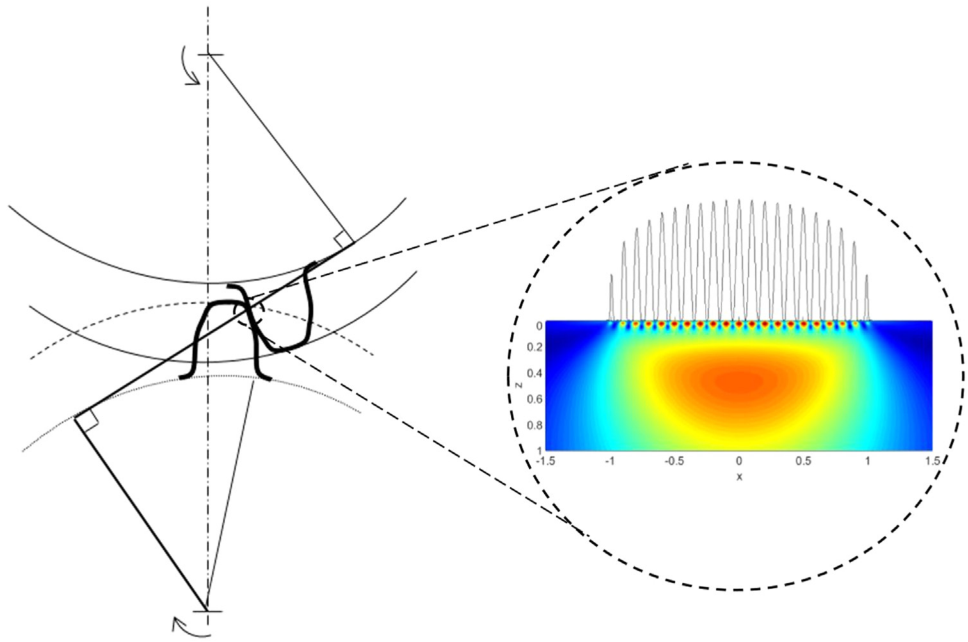

Fig. 1. Tooth contact between two spur gears. Enlarged view showing contact pressure distribution, surface and subsurface stresses of the contact. Surface stresses are mainly the results of the surface microgeometry of the gear tooth contact.

Thus, the life of a pair of meshing gears considering only the basic theory of Lundberg and Palmgren with 90% reliability is given by Eq. (A16) from Appendix A:

$$L_{10} = \left(\frac{W_m}{W}\right)^p \tag{2}$$

Where W is the tangential tooth load to the contact and $p = \frac{c+1-h}{2e}$ is the L&P life exponent, Appendix A.

Eqs. (1) and (2) apply to both spur and helical gears. However, for helical gears the calculation of the effective flank width f_e and the effective involute contact length l_e are different from spur gears (as explained in [11]) and are influenced by the helical angle, thus when redoing the algebra the equivalent of Eq. (1) is given by equation (19) from reference [11].

3. Separation of the surface from the subsurface survival

Although Eq. (2) can already calculate L10 life values for gears it cannot account explicitly for lubrication and particle contamination (thus, real surface failures). It is to say, this equation would be unable to distinguish failures due to micropitting or indentations and Hertzian pitting because it is a subsurface fatigue equation. Besides, it does not consider the influence of fatigue limit in the gear material neither for the surface nor the subsurface.

In rolling bearings some of these omissions have been remedied by introducing a fatigue limit stress from the steel and by multiplying the right-hand side of Eq. (2) by a global de-rating factor which considers lubrication quality, lubricant cleanliness and the fatigue limit of the material [7]. However, an even more flexible approach than this will be pursued in the current paper.

The approach follows the separation of survival for the surface and the subsurface, as introduced in [8] and followed in [9] for rolling bearings. The idea is that for the subsurface Hertzian contact theory, like Eq. (1), can continue to be used. But for the surface, more advanced models that can consider the real stresses of the lubrication conditions and other causes should be employed. In general in this region the tribological aspects of surface fatigue as shown in [14,15] have significant effect.

From Eq. (13) in reference [14] for a rolling contact, it has been shown that the probability of survival by dividing the contributions from the subsurface and the surface and assuming equal standardised Weibull slopes is given by:

$$Ln\left(\frac{1}{S}\right) = N^e\left[\overline{A}\int_{V_v}\frac{\langle\sigma_v - \sigma_{u.v}\rangle^c}{z^h}dV_v + \overline{B}\int_A\langle\sigma_s - \sigma_{u.s}\rangle^c dA\right] \tag{3}$$

From Appendix A, $L = uN$, thus from (3) solving for the life:

$$L = \frac{\left[Ln\left(\frac{1}{S}\right)\right]^{1/e}}{u}\left[\frac{1}{\overline{A}\int_{V_v}\frac{\langle\sigma_v - \sigma_{u.v}\rangle^c}{z^h}dV_v + \overline{B}\int_A\langle\sigma_s - \sigma_{u.s}\rangle^c dA}\right]^{1/e} \tag{4}$$

For a computer program based on stresses, Eq. (4) is recommended for the life estimation in gears. This would include consideration of misalignment, tolerances, profile accuracy and roughness. However, it is certainly laborious to implement such an equation in computer programs, especially because the surface integral would require a numerical method for mixed-lubrication surface stresses like the one described in [14,15]. Therefore, a first simplification of Eq. (4) would be to introduce a function (derived from the advanced surface distress model [14,15]) to account for the surface survival.

3.1. Surface survival

Following [14], a similar equation for surface survival can be curve-fitted. Thus, starting from:

$$I_s = \int_A\langle\sigma_s - \sigma_{u.s}\rangle^c dA \tag{5}$$

The derived equation (31) in [14] can also be used for gears, since it is based on a general rolling/sliding contact,

$$u^e I_s \approx f_1 exp\left[\frac{f_2}{(W/W_u)^{f_3}} + \frac{f_4}{(W/W_u)^{f_5}} + \cdots\right] \tag{6}$$

With particular values of the constants $f_1 \ldots f_n$ that would depend on the environmental factor $\eta = \eta_b\eta_c$ (lubrication and contamination) in gears, similar to rolling bearings. In Eq. (6) being W_u the fatigue load limit of the gear, defined in Appendix B.

Certainly, Eq. (6) would need to be calibrated for the particular case of gears using experiments or advanced numerical models [9,14,15,17] with different lubricant contamination cases and different lubrication qualities (Λ− ratios).

Therefore, the life Eq. (4) can now be written in the following from:

$$L = \frac{\left[Ln\left(\frac{1}{S}\right)\right]^{1/e}}{u}\left[\frac{1}{\overline{A}\int_{V_v}\frac{\langle\sigma_v-\sigma_{u.v}\rangle^c}{z^h}dV_v + f_1 exp\left[\frac{f_2}{(W/W_u)^{f_3}}+\frac{f_4}{(W/W_u)^{f_5}}+\cdots\right]}\right]^{1/e} \tag{7}$$

Eq. (7) is a possible representation of a stress-based model to be programmed in advanced computer programs for gears.

3.2. Towards a load-based model

For relatively simpler calculations, a load-based model is required, similar to Eq. (2). For this, one can start from equation (29) in [8]:

$$L_{10} = \frac{1}{\left[1+\frac{I_s}{Ln\left(\frac{1}{S}\right)}\left(\frac{W_m}{W}\right)^{pe}(a_u)^e\right]^{1/e}}a_u\left(\frac{W_m}{W}\right)^p \tag{8}$$

With,

$$a_u = \frac{\widehat{A}}{\left[1-\left(\Psi_{gear}\frac{W_u}{W}\right)^w\right]^{c/e}} \tag{9}$$

Where, Ψ_{gear} and $\widehat{A}$ are constants and $w = \frac{c-h+1}{2}$.

Eq. (8) can be rearranged to give,

$$L_{10} = \left[\frac{(a_u)^{-e}}{\left(\frac{W_m}{W}\right)^{ep}} + \frac{I_s}{Ln\left(\frac{1}{0.9}\right)}\right]^{-1/e} \tag{10}$$

Which clearly separates surface effects from subsurface effects. Now, normalising the surface integral,

$$\widetilde{I}_s = \frac{I_s}{Ln\left(\frac{1}{0.9}\right)}\left(\frac{W_m}{W}\right)^{ep} \tag{11}$$

One can then write Eq. (10) as,

$$L_{10} = \left[\frac{(a_u)^{-e}}{\left(\frac{W_m}{W}\right)^{ep}} + \frac{\widetilde{I}_s}{\left(\frac{W_m}{W}\right)^{ep}}\right]^{-1/e} \tag{12}$$

Eq. (12) becomes interesting, because it allows also for a separation of dynamic load into surface and subsurface influence. Since Eq. (12) needs to be calibrated with experimental L10 values for gears, it is conceivable to assume different values for W_m at the surface and the subsurface. This can be characterised by the use of different symbols.

$$L_{10} = \left[\frac{(a_u)^{-e}}{\left(\frac{W_{m.ss}}{W}\right)^{ep}} + \frac{\widetilde{I}_s}{\left(\frac{W_{m.s}}{W}\right)^{ep}}\right]^{-1/e} \tag{13}$$

In Eq. (13), the dynamic load capacity of the subsurface can simply be made equal to the previous value obtained in Eq. (1), thus $W_{m.ss} = W_m$.

To be consistent with the definition of dynamic load capacity, Appendix C derives an equation starting from stresses for the dynamic load capacity of the surface ($W_{m.s}$). Notice that this “rating” load for the surface reflects a sort of mean stress carried by the surface to attain a surface life L10 of 1 million pinion revolutions. Therefore, if the meshing gears have special design features (coatings, higher/lower surface hardness, etc.) that can extend or reduce the life of the surface, then this behaviour can be reflected by properly setting the constants in the definition of $W_{m.s}$, in the same way this can be done for $W_{m.ss}$ if there are special features for the subsurface (e.g. different material).

3.3. Surface fatigue indicator

One of the advantages of separating surface and subsurface survival in rolling contact fatigue is that it is possible to compare the inflicted damage of the surface respect to the total damage, in this way developing an indicator that can tell engineers what part of the rolling contact is enduring most of the fatigue damage. Besides, in a stress-based model this parameter can be output in a local basis along the surface. This is a very convenient way to develop possible mitigation strategies.

Therefore, following [8,9], thus from Eq. (13) by taking the denominators ratio between surface and the total (two added terms), one can define:

$$S_R = \frac{\widetilde{I}_s}{\widetilde{I}_s + (a_u)^{-e}} \tag{14}$$

This indicator, represents the relative fatigue damage taken by the surface in comparison to the total fatigue damage. If this parameter is close to 1, the majority of the fatigue damage is taken by the surface. Contrary, if this parameter is close to zero, then, most of the fatigue damage is taken by the subsurface. Notice, that this parameter by itself does not give direct indication that the surface or the subsurface is at imminent risk. Only when the calculated life is short then a possible risk of failure of one of the parts could be associated to S_R.

4. Application examples

The latest experimental results from Coy et al. are given in reference [12], however, before that the same authors used different parameter values for the same material. Table 1 gives a summary of the exponent and constant values for the dynamic load capacity of AISI 9310 gear steel.

Example 1. Spur gear data from references [10–13]. The main dimensions of the gear and pinion used in tests are summarised in Table 2.

The tests were done using super-refined, naphthenic, mineral-oil, with a viscosity of 73 cSt at 40 °C and 7.7 cSt at 100 °C, the oil contained 5% EP additive. The gear ratio was 1 and the speed was 10000 rpm. Coy et al. [13] report $\Lambda \approx 1.13$ (e.g. in rolling bearings, $\kappa = 1.17$, with $\kappa \approx \Lambda^{1.3}$). It would be considered a bit low to be full-film, it fits more to the mixed-lubrication regime.

With this data and following the detailed calculation procedure given in [10–13] the value for the meshing gear pair dynamic capacity can be calculated. Also applying the tangential transmitted load from the test [13], $W = 1617$ N. Now, here only the parameter values used in references [10,13] are applied.

$W_m = 16457[N]$, $L_{10} = 32.47$ [Mrevs] – as calculated from the reference's equations.

4.1. Use of the proposed model

The remaining data will be somehow assumed in analogy to rolling bearings, only for illustration purposes. Proper data would need to be

Table 1
Exponent and constant values considered in [10–13].

Parameter	Value [12]	Value [10,13]
e	2.5	3
c	23.2525	10.3333
h	2.7525	2.33333
w	10.75	4.5
p	4.3	1.5
k_1	5.28e8 [N-m]*	4.08e8 [N-m]

* taken from [13].

Table 2
Spur gear geometrical data from [10–13].

Parameter	Value
Tooth width in contact (flank width)	- 2.79 [mm]
Tooth width	6.4 [mm]
Number of teeth	28 [-]
Diametrical pitch	8 [-]
Circular pitch	9.975 [mm]
Whole depth	7.62 [mm]
Addendum	3.18 [mm]
Chordal tooth thickness ref.	4.85 [mm]
Pressure angle	20 [deg]
Pitch diameter	88.9 [mm]
Outside diameter	95.25 [mm]
Root fillet	1.02–1.52 [mm]
Measurement over pins	96.03–96.3 [mm]
Pin diameter	5.49 [mm]
Backlash reference	0.254 [mm]
Tip relief	0.01–0.015 [mm]

obtained from dedicated gear experiments. Consider the data of Table 3 for the application of Eq. (13).

The surface dynamic load capacity can be calculated with Eqs. (C2), (A14) and (A15), with data from Table 1 (last column) to give:

$$W_{m.s} = 53000 \quad N$$

The fatigue load limit can be calculated by using Eq. (B3), this gives:

$$W_u = 338.71 \quad N$$

In order to be able to apply Eq. (13), the value of $\widetilde{I}_s$ needs to be calculated first. Since at this point there is no yet a completely defined Eq. (6) for gears, the same diagram used for roller bearings in [8] will be used here as an example. However, since is gears the values of the exponents e and p are very different from roller bearings when normalising (application of Eq. (11)), the resulting diagram will be also different from the previous publication [8]. Fig. 2 shows the diagram adapted for the used exponents in gears.

As in [8] $\eta = \eta_b \eta_c$ and for gears the function η_b can be approximated to Λ by:

$$\eta_b \approx exp\left[\frac{0.3(\Lambda^{1.3}-4)}{(\Lambda^{1.3}-0.1)}\right] \tag{15}$$

Similar to [17] but here with constants adapted to gears.

Therefore, applying Eq. (13) for the current example with W/W_u= 4.77, and $\Lambda = 1.13$, thus from Eq. (15), $\eta = \eta_b \approx 0.48$,from Fig. 2, $\widetilde{I}_s = 8.13$.

Nominal life value calculated with Eq. (13) and Tables 1–3:

$$L_{10} = 15.92 \quad [\text{Mrevs}]$$

$$S_R = 0.96\,[-]$$

Notice that if arbitrarily $\widetilde{I}_s = 0$ is imposed, to remove any effect from the surface, the calculated life would go up to $L_{10} = 45.91$ [Mrevs], considering the effect of the fatigue limit stress the predicted life would be higher than the calculated value from [10,13]. However, in reality since $\widetilde{I}_s \neq 0$ the effects of the surface lower the predicted life.

4.2. Effect of load

Reducing the load to a half of its nominal value gives $L_{10} = 46.41$ [Mrevs], with $S_R = 0.99$ [-]. Increasing the load to twice its nominal value gives $L_{10} = 5.41$ [Mrevs], with $S_R = 0.88$ [-] maintaining all the rest of the nominal conditions unchanged.

4.3. Effect of the fatigue limit

Consider now one half of the nominal fatigue stress limit given in Table 3, thus $\tau_u = 180$ [MPa], and the rest of the nominal conditions, thus $W_u = 169.35N$ and $L_{10} = 14.40$ [Mrevs], with $S_R = 0.74$ [-]. The surface fatigue indicator clearly shows that the damage increases at the subsurface when the fatigue limit is reduced.

4.4. Effect of the lubrication quality

Consider now, $\Lambda = 0.2$, and the rest of the nominal conditions, thus $L_{10} = 8.36$ [Mrevs], with $S_R = 0.99$ [-]. And for $\Lambda = 3$, thus $L_{10} = 43.70$ [Mrevs], with $S_R = 0.14$ [-]. Here it is clear the effect that lubrication has not only on the life of the gears but also on the surface fatigue indicator S_R, when lubrication conditions worsen this indicator approaches 1, as in the case of, $\Lambda = 0.2$. On the contrary, when lubrication conditions improve, S_R approaches zero. Thus the surface fatigue indicator could indeed be part of a method to foresee micropitting risk.

4.5. Effect of the lubrication cleanliness

Assume lubricant cleanliness conditions of a gearbox, in bearings the contamination factor could be around $\eta_c \approx 0.1$. However, considering that gears have smaller number of contacts, perhaps the risk should be somehow reduced to $\eta_c \approx 0.2$ and all the other nominal conditions fixed. Then $L_{10} = 10.90$ [Mrevs], with $S_R = 0.99$ [-], the surface fatigue indicator clearly shows that all the damage is taken by the surface for this case.

4.6. Comparison with available endurance data

Krantz et al. [18] have published more recently endurance test data for the same NASA gears as references [10,12,13] and Table 2. They compared lives for ground finished gears with superfinished surfaces (polished using a process called barrelling) finding a considerable increase in fatigue life for the superfinished gears.

Here the summary from the test results of reference [18]. The mean value of the roughness for the standard ground gears is $R_a = 0.38\mu m$ and for the superfinished gears is $R_a = 0.07\mu m$. The gears were tested with a tooth load of 580 N/mm to give a Hertzian maximum pressure of 1.71 GPa at the pitch line. The test was conducted at 10000 rpm (46.5 m/s pitch-line velocity) and the calculated film thickness at the working temperature resulted in 0.54 μm. Therefore the $\Lambda-$ values for the standard ground and superfinished gears based on R_q are approximately 1.13 and 6.17, respectively. The measured L10 lives are $12x10^6$ and $46x10^6$ revolutions for the ground and superfinished gears. The reported failure modes are all fatigue but the analysed surfaces look different for the ground and the superfinished gears.

Comparing those data with the current model results, one can see that the standard ground gears correspond to the same example presented by references [10,12,13], it is to say, the nominal conditions for which the calculated life with the current model is:

$$L_{10} = 15.92 \quad [\text{Mrevs}]$$

$$S_R = 0.96\,[-]$$

Now for the superfinished case, one needs to run the example with $\Lambda = 6.17$ and all other conditions equal to nominal. This gives the following result:

Table 3
Assumed parameters to complete the use of Eq. (13).

Parameter	Value	Comment
τ_u	360 [MPa]	As for bearing steel [16]
Ψ_{gear}	0.15	As radial roller bearings
$\widehat{A}$	0.1	As rolling bearings
$\widetilde{I}_s$	same as fig. 7, [8]	
η_c	1	Very clean oil

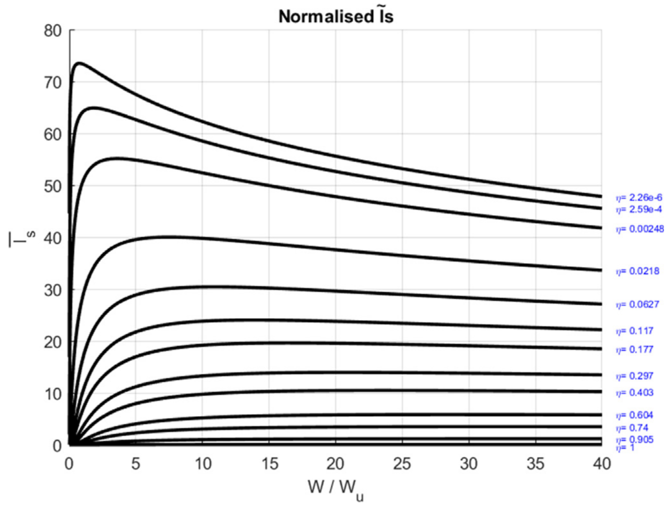

Fig. 2. Normalized $\widetilde{I}_s$ value (Eq. (11)) used in the examples, as adapted from roller bearings, reference [8] to gears.

$L_{10} = 43.70 \quad$ [Mrevs]

$S_R = 0.14$ [−]

The agreement with the measured values is reasonably good. Which shows that indeed the surface damage integral corresponding to roller bearings might also represent the case of spur gears.

The test data presented in [18] for the ground and superfinished gears have been digitized and replotted in Fig. 3 depicting a graphical comparison of the measured and calculated data.

Fig. 3 shows that the confidence intervals overlap a little and more failures would be required (especially for the superfinished variant) to reduce the uncertainty. In any case a clear differentiation in lives can be observed from the data. Table 4 shows the calculated Weibull maximum likelihood statistics associated to the data of Fig. 3.

The calculated lives are slightly overestimating the measurements, somehow more for the ground variant than from the superfinishing one. With the availability of further test, the model could be calibrated, following the methods already used in rolling bearings, [6]. However at this stage, it was felt that this was outside the scope of the paper which main objective was to give a first assessment to the basic theory without the interference of calibration factors. Considering this, as shown in Fig. 3, comparison between the test data and the calculated results using the methodology presented here shows a very good correlation.

Table 4
Weibull maximum likelihood statistics of gear endurance.

	Ground			Superfinish		
Confidence	L_{10}	L_{50}	β	L_{10}	L_{50}	β
50%	9.9	45.8	1.2	37.1	176.2	1.2
10%	5.6	33.5	1.0	15.6	116.7	0.8
90%	15.9	61.7	1.5	67.8	304.9	1.7

5. Discussion

The present concept model shows potential to include different surface failure modes in gears. More gear test results are needed to further validate the model, especially for other gear types different from spur. However, the current model with a surface damage function based on roller bearings seems to give good results for spur gears. For both parts (surface and subsurface), the effect of different gear materials would need to be considered. One aspect that adds especial interest is the consideration of any particular design feature that gears might have to mitigate possible surface failures, e.g. coatings, surface treatments, hardness, etc.

Although currently there are advanced numerical methods to predict micropitting in gears, considering the concurrent effects of surface fatigue and mild wear [14], it is clear that this advanced model cannot be used in every day calculations to estimate the life of gears. The method requires too many parameters, sometimes difficult to gather.

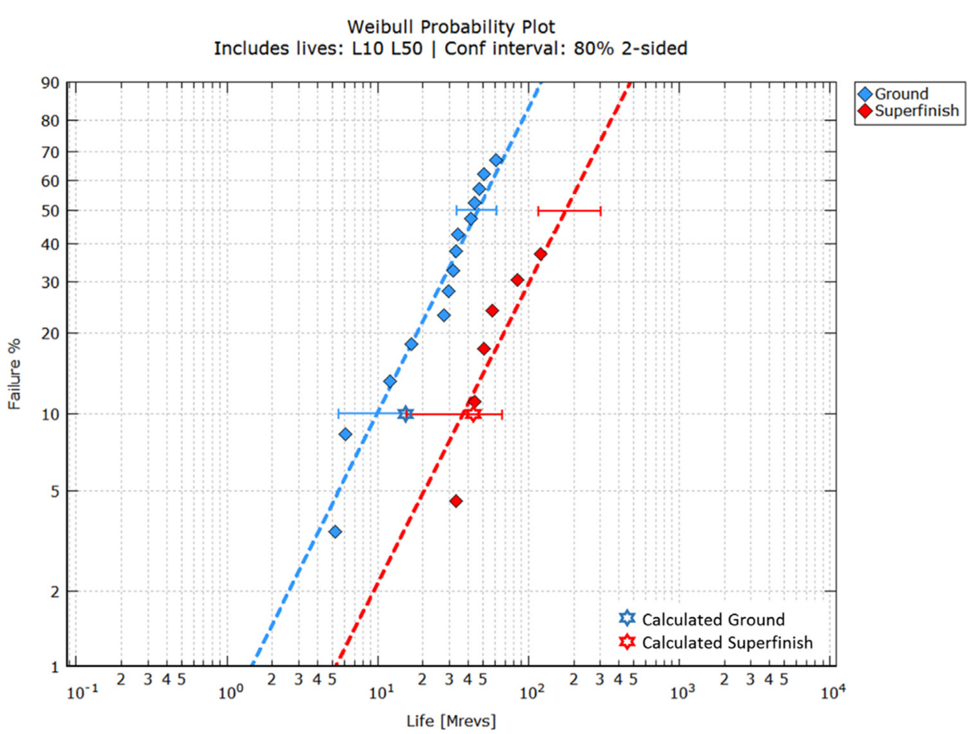

Fig. 3. Weibull plot for the two test cases, ground and superfinished gears, with calculated confidence intervals. The calculated values (stars) have been superimposed on the plot for comparison.

This is why it is needed a gear life calculation method more advanced than the current ones but less difficult to apply and more general than an advanced micropitting model.

Within the standards organisations there are currently ongoing efforts to provide design guidance for prevention of micropitting [3]. Improved prediction of gear micropitting would be helpful to better inform engineers about the most acceptable design criteria. For instance, increased sliding has been suggested to lead to increased micropitting, with the proposed mechanism [3,17] being that higher sliding leads to higher contact temperatures and thinner elastohydrodynamic film thicknesses, which in turn decreases the prevailing specific film thickness ($\Lambda-$ ratio) increasing in that way further the micropitting. This proposed mechanism requires further analysis, not only because sliding/rolling ratio (S) is generally considered to have minimal effect on the elastohydrodynamic film thickness (unless conditions close to pure sliding are encountered) and that it has been demonstrated in literature [15] the occurrence of micropitting in rolling bearings, which operate at much lower levels of sliding with typical sliding/rolling ratios (S) being around 0.05.

Therefore, the idea of having a general method for gear life prediction that can separate surface and subsurface becomes attractive, as already is happening in rolling bearings. This would allow more flexibility in identifying different failure modes, and also in a homologation of life calculation methods between gears and rolling bearings. Very useful, for instance, in gearbox design.

The current concept model has shown potential to give the correct influence of the different analysed parameters. For instance when the $\Lambda-$ ratio is reduced, the model clearly shows a remarkable reduction in calculated life and also the surface fatigue indicator rises. The same happens when lubricant cleanliness is compromised (by reducing the value of η_c). In the nominal conditions the model showed lower life than the prediction by references [10,13] but it is in agreement with measured values [18]. In any case, the nominal lubrication conditions described in the references do not seem to ensure complete full-film separation, the current model penalises already the calculated life for that. With superfinished surfaces the model is in agreement with measured values [18].

6. Conclusion

A concept model to separate the surface and subsurface survival for rolling contact fatigue initially developed for rolling bearings has now been adapted for gears. As examples of application, the effect of lubrication, contamination, fatigue limit and load have been shown.

The concept model can evolve into a predictive model once calibrated with the use of more experimental results on gears conducted under different application conditions, particularly the effect of material, lubrication, surface finish and particle contamination need to be accounted for.

In any case, even in a concept state the model can already give the correct values for spur gears and the gear material from references [10,12,13,18]. For the first time a gear life model shows an indicator of the fatigue damage distribution between the surface and the subsurface. Perhaps in the future this indicator can be part of a local surface distress (micropitting) criterion for gears.

From the analysis presented here, the following conclusions can be drawn:

1. The current approaches introduces separation of surface and subsurface survival. Therefore there could be significant gains by increasing flexibility in gear life modelling since more than one region is analysed. There is also the possibility to introduce surface failure modes in addition to the Hertzian rolling contact fatigue.
2. Since the surface effects are directly incorporated in the formulation. This approach allows for the exploitation of knowledge gained from the use of advanced tribological models. Non-tribological effects could be included in the future, like tooth root stresses.
3. The introduction of a surface dynamic load capacity in addition to the subsurface dynamic load capacity in gears is an innovation that would give flexibility to rate gears with special features, that clearly can influence the performance of the surface (e.g. surface treatments, coatings, surface finishing, etc.).
4. The current approach has shown that it can give an indication of the zone at higher fatigue risk in a gear population under given application conditions. This can help in the development of mitigation strategies.
5. Potentially it is possible to include different gear failure mechanisms, such as surface distress (micropitting), lubricant contamination, lubricant additive effects, wear-related stress concentrations, etc., capturing some deterministic aspects of bearing surface topography, material properties, and local lubrication conditions.
6. Particularly in the design of gearboxes, a homologation of life calculation methods between gears and rolling bearings would be extremely useful. Not only in the consideration of equal assumptions for bearings and gears but also in the consideration of common reliabilities.

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Appendix A. Meshing gear dynamic load capacity

The dynamic capacity of a gear tooth

It is the transmitted load by the tooth that for the contact surface will result in a L10 life of 1 million load cycles.

Starting from equation (37) of [5],

$$Ln\left(\frac{1}{S}\right) \propto \frac{\tau_o^c N^e}{z_o^h} V \tag{A1}$$

for gears the stressed volume is, $V = z_o f_e l_e$, thus.

$$Ln\left(\frac{1}{S}\right) \propto \frac{\tau_o^c N^e}{z_o^{h-1}} f_e l_e \tag{A2}$$

For gears, the Hertz problem can be simplified to a line contact geometry, thus the maximum Hertzian pressure p_o is related to the contact load Q by,

$$p_o = \frac{2Q}{\pi f_e b} \tag{A3}$$

where,

$$b = \sqrt{\frac{8QR}{\pi f_e E'}} \tag{A4}$$

Considering that τ_o is the orthogonal shear stress (τ_{xz}), thus $z_o = b/2$, and $\tau_o = \frac{p_o}{4}$ therefore, substituting into (A2),

$$Ln\left(\frac{1}{S}\right) \propto \frac{Q^{\frac{c}{2}}(\pi f_e E')^{\frac{h-1}{2}}(E')^{\frac{c}{2}} N^e f_e l_e}{\left(\frac{1}{2}\right)^{h-1}(8QR)^{\frac{h-1}{2}}\left(2\pi^{\frac{1}{2}} f_e^{\frac{1}{2}}\right)^{c}(8R)^{\frac{c}{2}}} \tag{A5}$$

Introducing the proportionality constant A_1 and grouping all constants and the elastic constants in one constant k, one has.

$$Ln\left(\frac{1}{S}\right) = kQ^{\frac{c-h+1}{2}} f_e^{\frac{h+1-c}{2}} R^{\frac{1-h-c}{2}} N^e l_e \tag{A6}$$

with

$$k = A_1\left(\frac{\pi E'}{8}\right)^{\frac{c+h-1}{2}} \pi^{-c} 2^{h-1-c} \tag{A7}$$

From Eq. (A7) the load can be calculated, replacing $N = uL$ being u a conversion factor from load cycles to gear revolutions, normally $u \approx 1$ cycle/revolution.

$$Q = \left(\frac{k}{Ln\left(\frac{1}{S}\right)}\right)^{\frac{2}{h-1-c}} f_e^{\frac{h+1-c}{h-1-c}} R^{\frac{1-h-c}{h-1-c}} l_e^{\frac{2}{h-1-c}} (uL)^{\frac{2e}{h-1-e}} \tag{A8}$$

Now, for the calculation of the dynamic load capacity, take L = 1 (M revs) and S = 0.9, thus the dynamic load capacity for any gear is given by,

$$Q_c = \left(\frac{k}{Ln\left(\frac{1}{0.9}\right)}\right)^{\frac{2}{h-1-c}} f_e^{\frac{h+1-c}{h-1-c}} R^{\frac{1-h-c}{h-1-c}} l_e^{\frac{2}{h-1-c}} u^{\frac{2e}{h-1-e}} \tag{A9}$$

with $k_0 = \left(\frac{k}{Ln\left(\frac{1}{0.9}\right)}\right)^{\frac{2}{h-1-c}}$ and from reference [10], equation (42) it can be concluded that, $k_1 = k_0 \pi^{\frac{-2}{c-h+1}}$. Coy et al. [10] also give values of k_1.

Now, for a spur gear, f_e is the flank width and l_e is the total length of the involute zone of contact. For an helical gear, as explained in [11] these quantities depend also on the helical angle and equations are given in the reference. For both gear types $u \approx 1$.

Now, from Coy et al. [10] by following only the procedure of Lundberg and Palmgren [5] they give the life of the individual gear tooth as,

$$L_{10} = \left(\frac{Q_c}{Q}\right)^p \tag{A10}$$

with $p = \frac{c+1-h}{2e}$

The dynamic capacity of a gear

The method of probabilities is followed to obtain composed lives, thus, for a gear with many teeth,

$$\left(\frac{1}{L}\right)^e = \left(\frac{1}{L_1}\right)^e + \left(\frac{1}{L_2}\right)^e + \dots + \left(\frac{1}{L_n}\right)^e \tag{A11}$$

with every tooth equal,

$$\left(\frac{1}{L}\right)^e = n\left(\frac{1}{L_i}\right)^c \tag{A12}$$

Finally, substituting (A10) into (A12)

$$L_{10} = \left(\frac{Q_c}{n^{\frac{1}{pe}} Q}\right)^p \tag{A13}$$

The dynamic capacity of a pair of meshing gears

Applying the same probability concepts of Eq. (A11), and following Coy et al. [10] it is possible to find the dynamic capacity of a pair of meshing gears where subscript 1 refers to pinion and subscript 2 to gear. Thus, the dynamic capacity of the mesh of the transmitted tangential load that might be carried for 1 million revolutions of the pinion with a 90% reliability is given by:

$$W_m = Q_m cos(\varphi) \tag{A14}$$

Where,

$$Q_m = \left[N_1\left(\frac{1}{Q_{c1}}\right)^w + N_2\left(\frac{1}{Q_{c2}}\right)^w\right]^{-1/w} \tag{A15}$$

With Q_{c1} given from Eq. (A9) for the pinion and Q_{c2} given from Eq. (A9) for the gear, and $w = \frac{c-h+1}{2}$.

Finally using only Lundberg and Palmgren concepts, the life of a gear pair is given by

$$L_{10} = \left(\frac{W_m}{W}\right)^p \tag{A16}$$

Appendix B. Fatigue load limit in gears

The particular value of the fatigue stress limit for gear materials will depend on the material itself, however, as an example one could start by assuming the same value as for bearings [16], the actual fatigue stress limit can also be assumed to be zero, in which case $W_u = 0$ and the load in Eq. (6) would need to be normalized with the dynamic load capacity rather than the fatigue load limit.

Assuming that there is an orthogonal shear stress τ_{xz} value below which material fatigue does not develop, call it τ_u. Thus, the gear normal load needed to produce such an orthogonal shear stress following Eq. (A3) is,

$$Q_u = \frac{\pi f_e b \tau_u}{2} \tag{B1}$$

Now the limiting transmitted load in the gear pair is,

$$W_u = Q_u \cos(\varphi)\cos(\psi) \tag{B2}$$

Being φ the pressure angle and ψ the helical angle. Therefore:

$$W_u = \frac{\pi f_e b \tau_u}{2}\cos(\varphi)\cos(\psi) \tag{B3}$$

Appendix C. Dynamic load capacity for the surface in gears

The dynamic capacity for the surface of a gear tooth

It is the transmitted load by the tooth surface that will result in a L10 surface life of 1 million load cycles. Thus starting from Eq. (A2):

$$Ln\left(\frac{1}{S}\right) \propto \frac{\tau_o^c N^e}{z_o^{h-1}} f_e l_e \tag{C1}$$

Now, at the surface one can neglect the stress depth z_o, thus making $h = 1$, and also considering that $\tau_o = Tp_o$, where T is a constant. However, different from the subsurface, at the surface this constant is unknown and can be simply carried away with the constant terms, to give a surface dynamic capacity for a gear tooth of,

$$Q_{c.s} = \left(\frac{k'}{Ln\left(\frac{1}{0.9}\right)}\right)^{-2/c} f_e^{\frac{-c}{2-c}} R l_e^{\frac{-c}{2}} \tag{C2}$$

with $k'_0 = \left(\frac{k'}{Ln\left(\frac{1}{0.9}\right)}\right)^{\frac{-2}{c}}$ and from reference [10], equation (42) it can be concluded that, $k'_1 = k'_0 \pi^{\frac{-2}{c}}$.

Notice that $k_1{}'$ can be different from k_1 of Appendix A if different performance is allocated to the surface, however in the examples of this article the same values were used.

After applying Eq. (C2) Eqs. (A14) and (A15) can be used to calculate the surface dynamic load capacity of the pair of meshing gears.

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